Physics-Informed Machine Learning

Constraint-Preserving PINNs for *Lindblad* Quantum Dynamics

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MOTIVATION AND BACKGROUND

Open quantum systems evolve continuously. Most methods discretize. We need the network itself to *be* the solution.

The physical setting

Qubits in realistic environments lose coherence over time. Modeling this dynamics is central to quantum error correction and device characterization.

The computational gap

Classical ODE solvers give a discretized trajectory. A neural network representation gives a *continuous function of time* that satisfies the governing equation everywhere.

The open question

Standard approaches cannot guarantee that the network output is a *valid quantum state* at every point. The challenge is to model continuous quantum dynamics without leaving the space of physical states.

SIXTY YEARS IN ONE SLIDE

Neural networks learn functions. Physics-informed neural networks learn to satisfy equations.

1958	1986	1989	2019
Perceptron Rosenblatt	Backprop Rumelhart et al.	Universal Approximation	PINNs Raissi et al.

Neural networks are universal function approximators. PINNs exploit automatic differentiation to evaluate PDE residuals through the network itself — replacing labeled data with the governing equation.

Data-driven learning

$L = \|NN_\theta(x_i) - y_i\|^2$

fit predictions to observed input-output pairs

Physics-informed learning

$L = \|\partial u_\theta/\partial t - \alpha\partial^2 u_\theta/\partial x^2\|^2$

fit predictions to a known PDE, e.g. the heat equation

The network returns one value each time it is evaluated, but because the inputs vary continuously, the trained model represents a continuous function.

WHAT WE'RE TRYING TO DO

Teach a neural network to represent the *quantum state* of an open system as it evolves in time.

The state

The state of an open quantum system is a matrix `ρ(t)` — the *density matrix*. Diagonal entries are probabilities; off-diagonals encode coherence.

The equation it obeys

Its evolution is governed by the *Lindblad master equation* — a PDE for ρ(t). H drives coherent evolution; L is the dissipator coupling to the environment.

```
dρ/dt = -i[H, ρ] + LρL† - ½{L†L, ρ}
```

The catch

To be a valid quantum state, ρ must satisfy three properties exactly: Hermitian (`ρ=ρ†`), positive (`ρ≥0`), and unit trace (`Tr(ρ)=1`). A raw network output satisfies none.

A common workaround is to *penalize* violations in the loss and hope they are small at convergence. That still leaves validity as something the model must learn rather than something the output guarantees.

THE MISMATCH

Traditional neural networks cannot guarantee valid density-matrix output.

An MLP mapping $t \mapsto NN_\theta(t) \in \mathbb{C}^{2\times 2}$ emits eight unconstrained real values. None of ρ's defining properties hold.

I. Hermiticity

ρ must equal its conjugate transpose. Off-diagonals must be complex conjugates of each other. The network emits ρ_{01} and ρ_{10} independently — nothing ties them together.

ρ_{01} = 0.3+0.4i
ρ_{10} = 0.1-0.2i

not conjugate → not Hermitian

II. Positivity

Eigenvalues of ρ are probabilities — they cannot be negative. A raw matrix has no spectral constraint; eigenvalues can land anywhere, including below zero.

λ_1 = 1.12
λ_2 = -0.12

negative probability → unphysical

III. Unit trace

Diagonal entries are basis probabilities — they must sum to exactly 1. The network's two diagonal outputs are unconstrained real numbers.

ρ_{00} = 0.62
ρ_{11} = 0.41

Tr(ρ) = 1.03 → over 100%

The standard fix: add penalty terms λ_H, λ_T, λ_P to the loss and hope they shrink at convergence. In practice, competing gradient magnitudes destabilize training and violations persist.

THE KEY IDEA

$$\rho(t) = \frac{A(t)\,A^{\dagger}(t)}{\mathrm{Tr}[\,A(t)\,A^{\dagger}(t)\,]}$$

Hermitian

$(AA^{\dagger})^{\dagger} = AA^{\dagger}$. Always.

Positive semidefinite

$\langle x, AA^{\dagger}x\rangle = \|A^{\dagger}x\|^2 \geq 0$.

Unit trace

Division by $\mathrm{Tr}(AA^{\dagger}) > 0$ makes it exactly one.

The network outputs $A(t)$. ρ is built from A. Every output is a valid quantum state — no penalties, no tuning.

HARD CONSTRAINT FIRST

The output parameterization makes every predicted quantum state valid.

The neural network does not predict p_0, p_1, and c directly. It predicts four unconstrained real functions, then a trace-normalized Gram matrix converts them into a valid density matrix.

Target structure

$$\rho_\theta(t) = \begin{vmatrix} p_{0,\theta}(t) & c_\theta(t) \\ c_\theta^*(t) & p_{1,\theta}(t) \end{vmatrix}$$

A single-qubit density matrix has populations on the diagonal and coherence off the diagonal.

→

Neural output

$$NN_\theta(t) = [\alpha_1(t), \alpha_2(t), \beta(t), \gamma(t)]$$

$$A_\theta(t) = \begin{vmatrix} \exp(\alpha_1(t)) & 0 \\ \beta(t)+i\gamma(t) & \exp(\alpha_2(t)) \end{vmatrix}$$

The network outputs hidden construction variables, not physical density-matrix entries.

→

Valid density matrix

$$\rho_\theta(t) = A_\theta(t)A_\theta^\dagger(t) \;/\; \mathrm{Tr}(A_\theta(t)A_\theta^\dagger(t))$$

$$\rho_\theta^\dagger = \rho_\theta \quad \rho_\theta \geq 0 \quad \mathrm{Tr}(\rho_\theta) = 1$$

The output is Hermitian, positive semidefinite, and trace-one by construction.

PHYSICS LOSS SECOND

After the output is valid, the loss checks whether it obeys Lindblad dynamics.

$$\mathcal{R}_\theta(t) = d\rho_\theta/dt - \mathcal{L}[\rho_\theta]$$

residual = learned time derivative − Lindblad generator

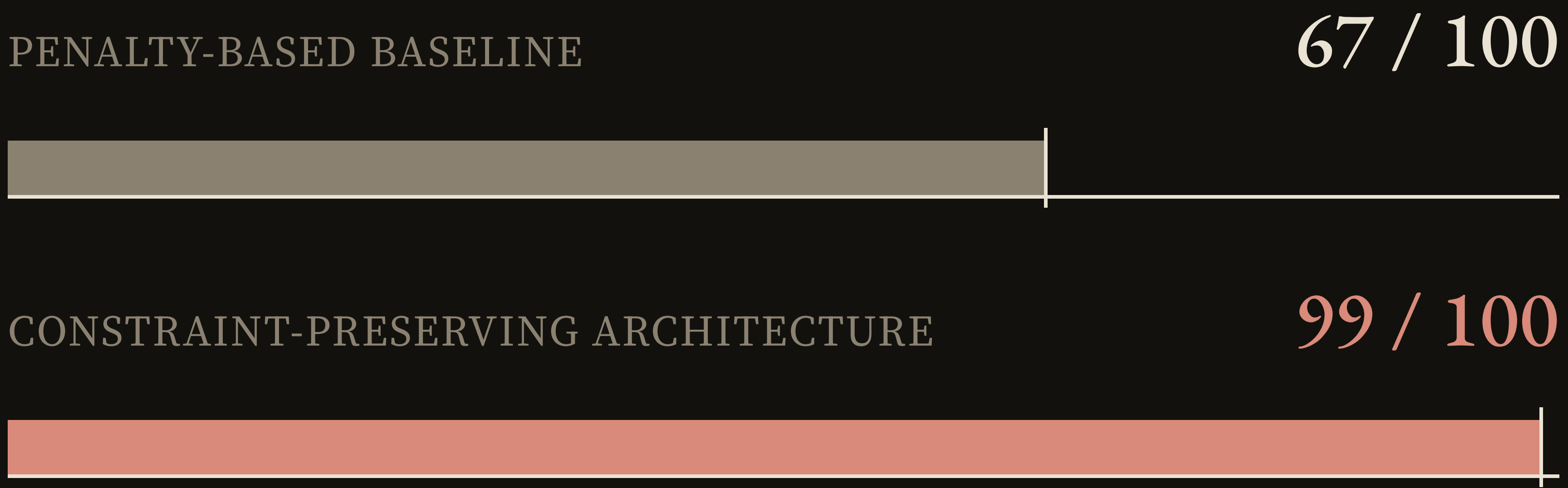
$$\mathcal{L}_{phys} = \| \, d\rho_\theta/dt - \mathcal{L}[\rho_\theta] \, \|^2_F$$

No Hermiticity, trace, or positivity penalty terms are needed.

The parameterization first maps the neural output into the valid quantum-state space, then evaluates whether that valid state obeys Lindblad dynamics.

RESULT: ARCHITECTURE REDUCED FAILURE MODES ACROSS SEEDS

The strongest result is the failure-rate collapse.



The constraint-preserving model produced 32 additional runs that satisfied all validity gates across identical 5000-epoch training runs.

What counted as success

A run only counted if it satisfied the PDE residual gate and stayed inside the density-matrix validity requirements.

Criterion	$\lvert \mathrm{Tr}(\rho) - 1 \rvert < 10^{-12}$
Criterion	$\lambda_{min}(\rho) \geq -10^{-10}$
Criterion	Final PDE residual $< 10^{-5}$
Takeaway	The architecture made successful training much more repeatable across random seeds.

Table 2 · 100 seeds

RESULTS SUMMARY

Does it train, stay valid, and learn the physics?

Building quantum constraints into the *representation* improved reliability, preserved validity for all `t`, and still learned the dephasing dynamics.

The core evidence is simulation-based: repeated training reliability, time-resolved constraint preservation, and learned dephasing behavior. Hardware validation remains future work.

Result 1

Reliability across seeds

Does the architecture train more reliably than the penalty baseline?

99 vs 67

successes out of 100

Result 2

Physical validity for all `t`

Does the output remain a valid density matrix across the trajectory?

$\leq 10^{-15}$

trace error scale

Result 3

Learned physics behavior

Does the learned trajectory still follow pure dephasing behavior?

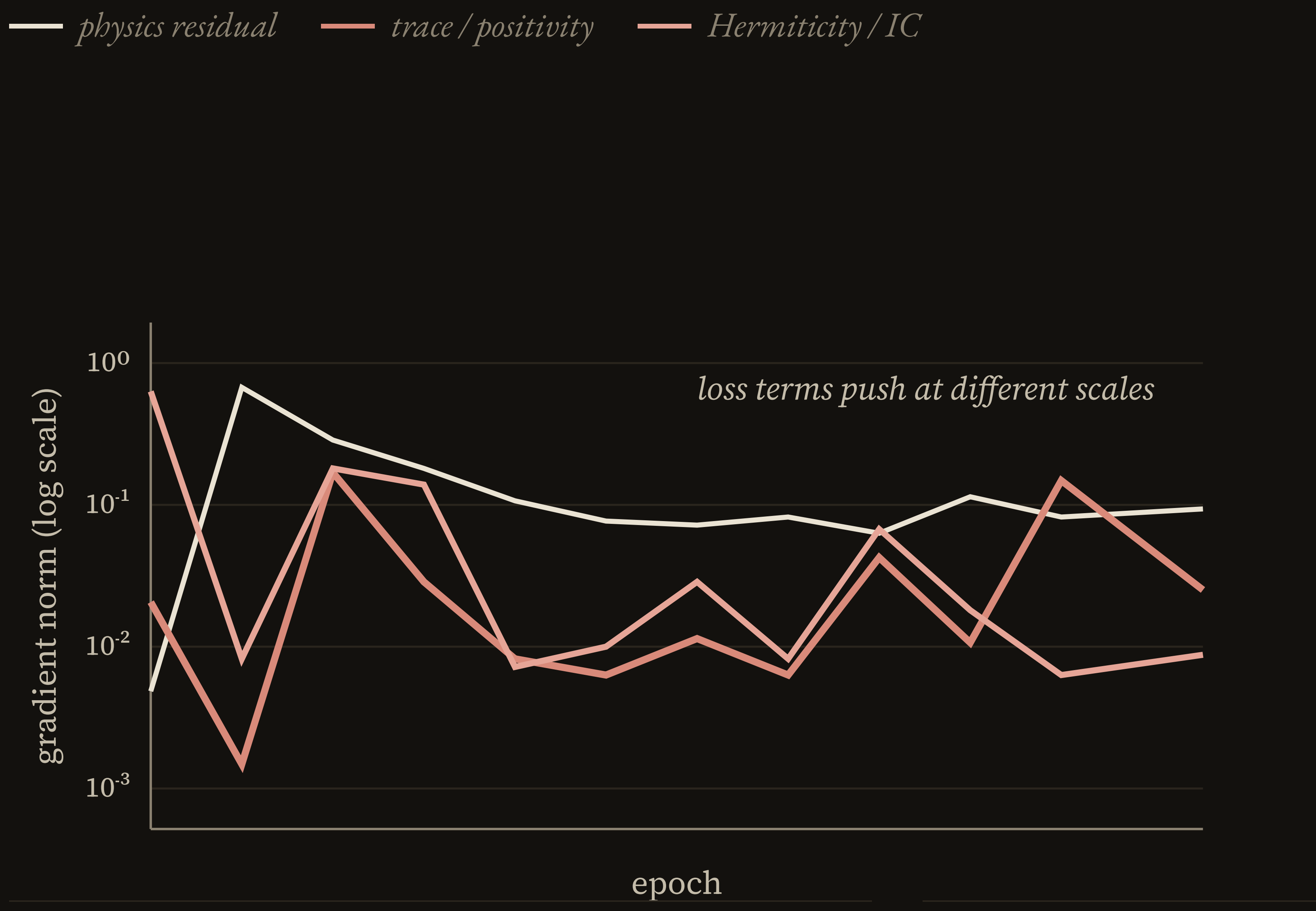
dephasing

behavior recovered

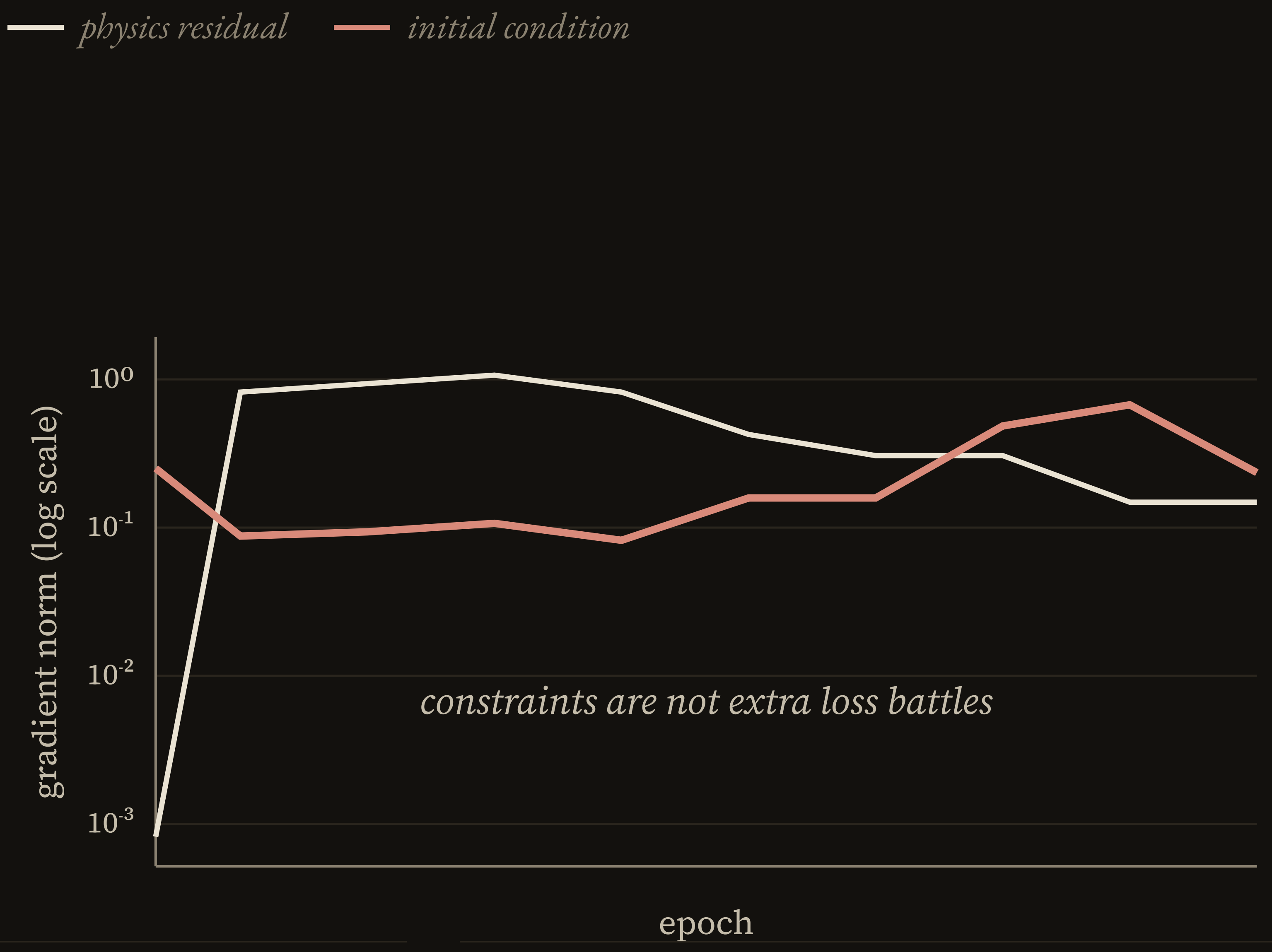
RESULT: PENALTY TERMS CREATE GRADIENT COMPETITION

The baseline must balance several competing gradients.

Penalty-based baseline



Constraint-preserving model



Baseline: physics, Hermiticity, trace, positivity, and initial-condition terms all compete during training.

Architecture: Hermiticity, trace, and positivity are handled before the loss is even computed.

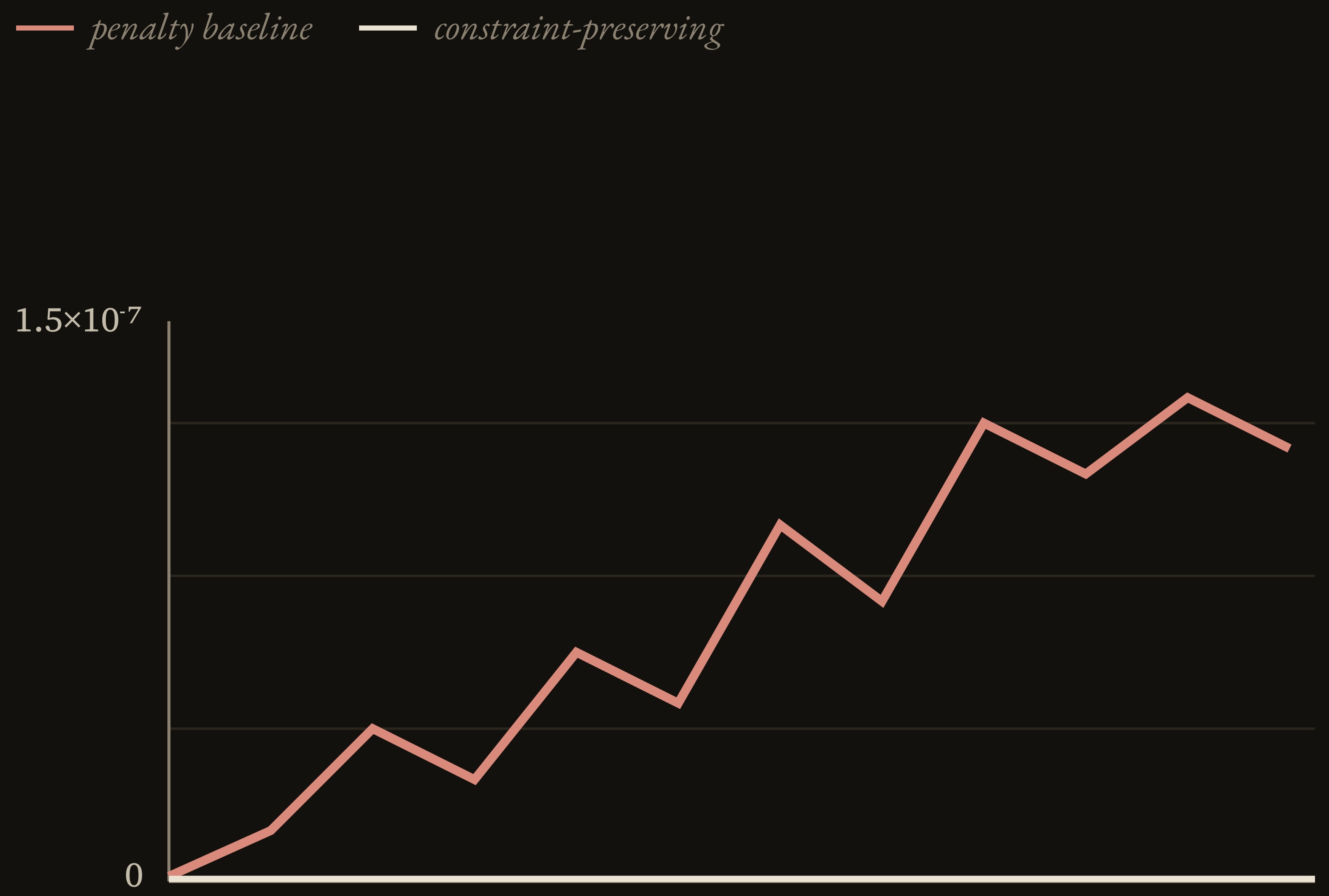
Implication: this explains why reliability improves, not merely that the final score is higher.

Figure 7 · gradient norms by loss term

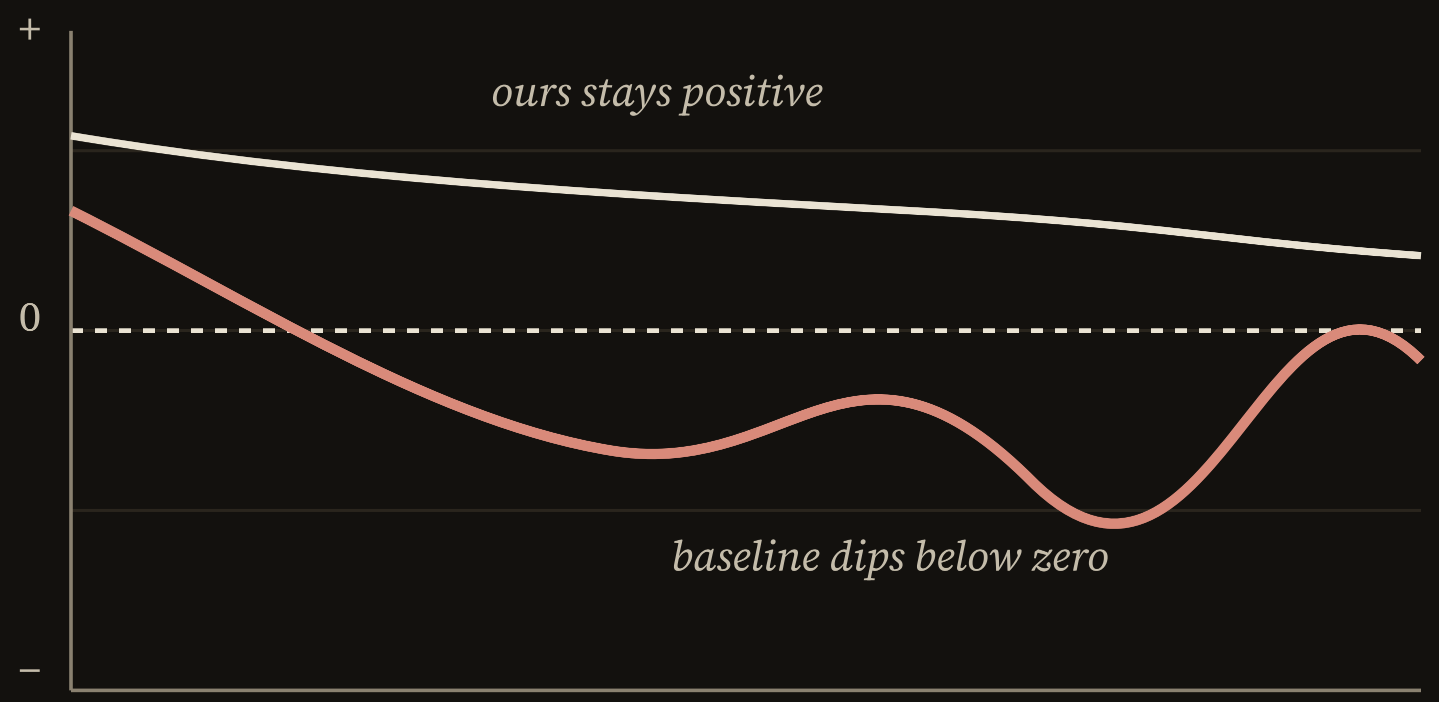
RESULT: VALIDITY HOLDS ACROSS THE LEARNED TRAJECTORY

We evaluate physical validity for *every* t rather than relying on an aggregate score.

Trace error $|\mathrm{Tr}(\rho)-1|$



Minimum eigenvalue $\lambda_{\min}(\rho)$



Mechanism: Gram normalization forces $\mathrm{Tr}(\rho)=1$, so trace error stays at numerical precision.

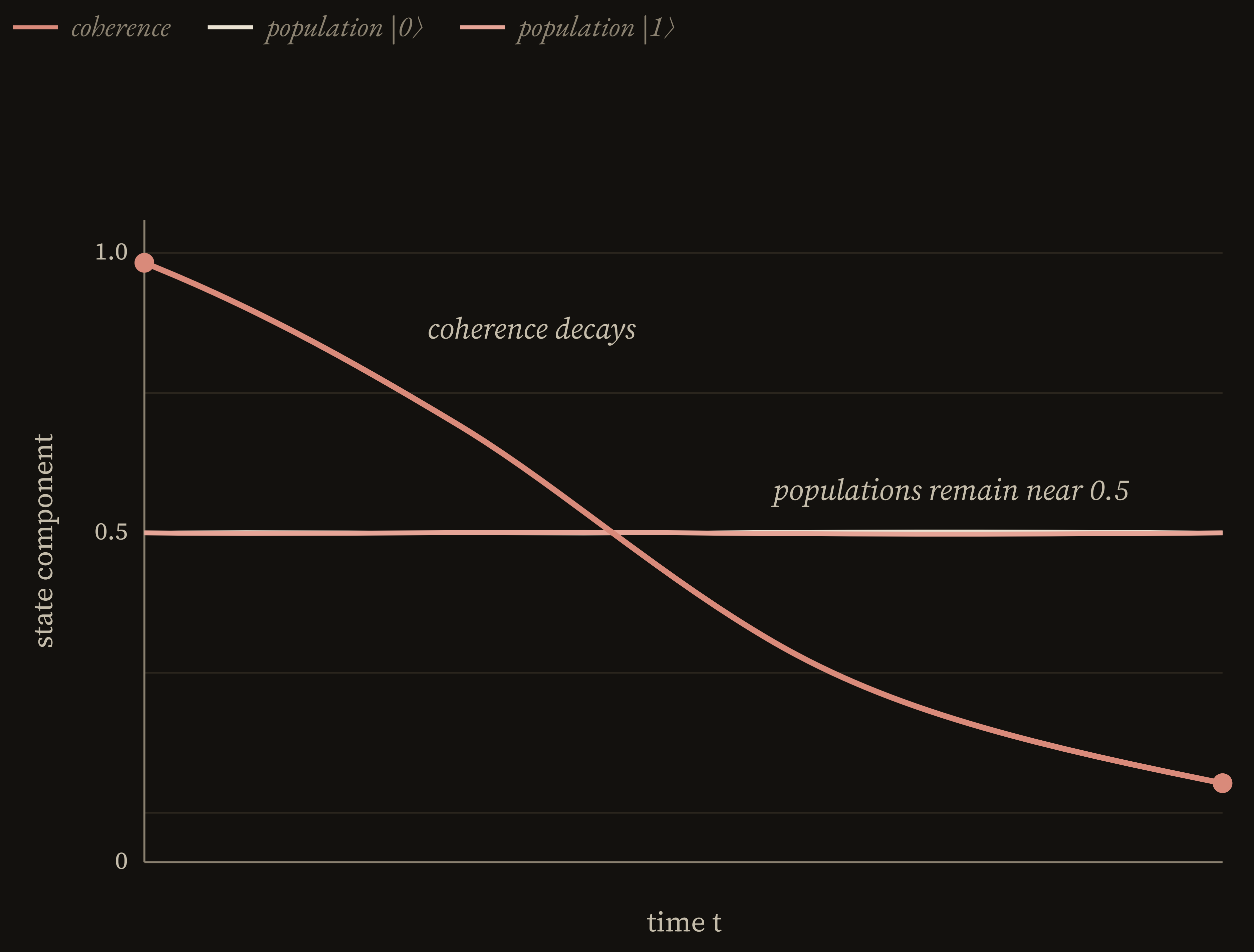
Observation: Baseline drifts around 10^{-7}; constraint-preserving output stays near machine precision.

Interpretation: The plot supports the hypothesis because the constraint is satisfied by representation, not by tuning λ weights.

Figure 10 · trace & eigenvalue comparison

RESULT: THE MODEL LEARNS DEPHASING BEHAVIOR

Under pure dephasing, populations stay fixed while coherence decays.



What the learned trajectory shows

The model reproduces the qualitative signature of pure dephasing while staying inside the density-matrix constraints.

`Populations`	remain essentially fixed over time
`Coherence`	decays as the off-diagonal terms vanish
`Meaning`	validity is built in, and the learned dynamics still match the physics

Figure 11 · learned quantum decoherence dynamics

FROM VALID MODELING TO CONTROLLED QUANTUM DYNAMICS

Next: validate, scale, then tune coherence.

Clean hardware validation

Prepare the same $|+\rangle$ initial state on hardware, run the dephasing protocol, and compare like-for-like trajectories.

The current attempt exposed a protocol mismatch, so hardware remains future work rather than evidence for the main claim.

Multi-qubit channels

Extend the constraint-preserving parameterization beyond a single-qubit density matrix to larger systems and quantum channels.

The natural next constraint is complete positivity and trace preservation for channel-level dynamics.

Quantum Tuning

Use SNAP-ADS to predict decoherence before it appears in the quantum system.

Then tune the oscillator preemptively to keep phase dynamics coherent.

PREPRINTS AND SUBMISSIONS

Two works in or approaching preprint.

Constraint-Preserving PINNs for Lindblad Quantum Dynamics

Nathan Hangen · Wright State University · 2026

Introduces a Gram-matrix parameterization that guarantees physical validity of density-matrix PINNs by construction. 99% success rate versus 67% for penalty-based baseline across 100 trials.

Preprint (arXiv, submitted)

`github.com/nhangen/constraint-preserving-pinns`

Gauge-Invariant Time Perception and the Minimal Conditions for Interior Sensitivity

Nathan Hangen · Wright State University · 2026

Applies reparameterization invariance from physics to mathematical models of human time perception. Shows that memoryless gauge-invariant functionals collapse to endpoint readout; memory is necessary but not sufficient for interior sensitivity.

Manuscript in preparation (preprint planned)

`github.com/nhangen/gauge-invariant-time-perception`

CITED SOURCES

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WITH THANKS

Acknowledgements

Wright State University

Department of Physics

Dr. Ajani L. Ross
Senior project mentor and advisor

Amit Sharma
Program oversight

Manuscript feedback

Soumyadip Sarkar
manuscript feedback; arXiv quant-ph endorsement

arXiv endorsement

D. P. Kingma
arXiv cs.LG endorsement

This work was conducted independently.